

TradeLoop — Technical Architecture

Backend & build plan · validate thin, then scale

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How TradeLoop goes from a static demo to a real, multi-tenant SaaS that's efficient and scalable. Written for Logan + Liam (technical). The headline call is in §1 — **validate with a thin stack before building custom**.

1. The strategic call: validate thin, then build custom

Don't build the full custom platform first. The loop is unproven with paying clients, and a from-scratch build is 3+ months. Two phases:

- **Phase A — "Concierge stack" (weeks, not months).** Wire the loop together with off-the-shelf parts and deliver it *done-for-you* to the first 3–5 clients (the \$1,200 + \$200/mo offer). Proves the loop converts, banks cash, and teaches you exactly what to build. Stack: **Twilio (or NZ TNZ/MessageBird) + n8n/Make + Xero API + Claude + a Google Sheet/Airtable**. You operate it; the client just sees results.
- **Phase B — Custom product.** Once the loop is proven and you have ~5 paying clients and real usage data, build the multi-tenant product below for scale and margin.

This sequence is the difference between cooking and stalling. Everything past §3 is Phase B.

2. Phase A concierge stack (build this week)

Need	Tool	Why
Inbound number + missed-call detection + SMS	Twilio (global) or TNZ / MessageBird (NZ-native, better deliverability)	Forward the tradie's number; fire a webhook on missed call
Orchestration	n8n (self-host, cheap) or Make	Webhook → AI → SMS → log, no code
The reply brain	Claude (Anthropic API)	Drafts the SMS in the tradie's voice, detects booking intent
Invoicing	Xero API	Draft + send + reconcile
Lead store + digest	Airtable / Google Sheet	Pipeline + the morning digest, zero build
Notifications	Slack / email	Owner approves from their phone

Cost to run per client ≈ SMS + a few cents of LLM + number rental — comfortably inside \$200/mo.

3. Phase B — recommended production stack

Keep it on the infra you already use (Vercel) and lean serverless — the workload is webhook-driven and bursty, which is exactly what serverless is cheap and scalable for.

- **App + API:** Next.js (App Router) on **Vercel**, TypeScript. API routes / route handlers for webhooks.
 - **DB:** Postgres (**Neon** or Vercel Postgres) + **Prisma** ORM. Row-level multi-tenancy.
 - **Queue + schedule:** **Upstash QStash** (HTTP queue + cron) for sequences, digest, retries — serverless-friendly, no always-on worker.
 - **SMS/voice:** **Twilio** + an NZ sender (**TNZ/MessageBird**) for deliverability.
 - **Accounting:** **Xero** OAuth2 (MYOB later).
 - **AI:** **Claude** (Anthropic) with tool-use for booking/quote actions.
 - **Auth:** Auth.js or Clerk. **Billing:** **Stripe** (NZD, GST).
 - **Email:** Resend or Postmark. **Errors:** Sentry. **Analytics:** Plausible (privacy-friendly).
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4. Data model (multi-tenant — every row carries `businessId`)

```
Business    id, name, trade, phone, xeroTokenId, tone, marginTarget, invoiceTerms, plan, status
User        id, businessId, name, email, role
PhoneNumber id, businessId, e164, provider
Contact     id, businessId, name, phone, email, lastSeenAt
Call        id, businessId, contactId, direction, status(missed/answered), at
Message     id, businessId, contactId, channel(sms), direction, body, at      // the thread
Lead        id, businessId, contactId, stage(hot/warm/reactivation/won), valueEst, source
Job         id, businessId, leadId, status, completedAt
Quote       id, businessId, jobId, quoted, estCost, marginPct, flagged
Invoice     id, businessId, jobId, xeroInvoiceId, amount, status, sentAt, paidAt
Sequence    id, businessId, type(revival/followup), name
SeqStep     id, sequenceId, offsetDays, channel, template
Enrolment   id, businessId, contactId, sequenceId, step, nextRunAt, status
DigestItem  id, businessId, type, payload, status(pending/approved/denied), date
Event       id, businessId, kind, payload, at      // audit log of everything the system did
Setting     id, businessId, key, value
```

5. Core pipelines (the product)

A. Missed call → booked job (the magic)

```
Twilio webhook (missed call)
→ upsert Contact, create Call(status=missed)
→ enqueue respondToMissedCall(businessId, contactId)
  → load Business context (tone, services, pricing bands, booking rules)
  → Claude drafts SMS in the owner's voice (tools: proposeBooking, quoteRange)
  → send SMS via provider; store Message(out)
inbound SMS webhook → store Message(in) → Claude continues thread
  → on booking intent: create Lead(stage=hot) + DigestItem(type=booking)
  → optional: write to calendar
```

B. Job done → paid

```
Job.completedAt set
→ draft Invoice from parts+labour → Xero draft → DigestItem(type=invoice)
→ owner approves in digest → send via Xero + payment link
→ Xero payment webhook → Invoice.paidAt → reconcile → Event
```

C. Lead revival — QStash cron enrolls cold quotes / past customers, fires timed SeqSteps (SMS/email), exits on reply.

D. Pricing guard — on Quote create, $\text{marginPct} = (\text{quoted} - \text{estCost}) / \text{quoted}$; if $< \text{marginTarget}$ → flag + DigestItem.

E. Daily digest — QStash cron 6:00 NZT → assemble pending DigestItems → push to owner (app + SMS/email). Owner taps approve → actions fire.

6. The AI layer (do this carefully — it's the product)

- **Per-business system prompt:** voice/tone, services, pricing bands, service area, booking rules, what to never promise.
- **Tool-use** for real actions (proposeBooking, createQuote, escalateToHuman) — don't let the model free-text commitments.
- **Guardrails:** confidence threshold → hand to human; always allow the owner to take over a thread; never quote outside configured bands.
- **Log every message** for trust, debugging, and tuning. Honest marketing — easy human handoff, no "fully human" overclaim.

7. Security & NZ compliance

- Privacy Act 2020: data minimisation, access/delete, encrypted Xero tokens, secrets in Vercel env (never in repo).
- **UEMA 2007:** honour STOP/opt-out on every outbound message; record consent basis.
- SMS sender compliance + NZ number provisioning. Audit log (Event) of all automated actions.

8. Cost-to-serve & margin (per client / month)

- SMS: $\sim \$0.10 - 0.25 / \text{msg} \times \sim 80 - 200 \text{ msgs} \approx \text{\$10-40}$
- LLM: cents per conversation $\approx \text{\$2-10}$
- Number rental + infra share $\approx \text{\$3-8}$
- **Total $\approx \text{\$20-55/client}$.** At $\$149 - 249 / \text{mo}$ that's $\sim 75 - 85\%$ **gross margin**. Scales well; watch SMS volume on heavy after-hours tenants (cap/upsell).

9. Scaling — what breaks and the fix

Clients	What breaks	Fix
1–5	Your time (concierge setup)	Templatise onboarding; checklist + scripts
5–25	Manual setup, support load	Self-serve onboarding wizard; SA pod runs support
25–100	SMS/LLM cost variance, Xero rate limits	Queues, batching, per-tenant usage caps, caching
100+	Multi-region, voice volume, reporting	Dedicated number pools, voice tier, data warehouse

10. Build phases & rough effort

- **Phase A concierge:** ~1 week (Liam) — proves the loop, first revenue.
- **Phase B MVP:** ~4–6 weeks — single→multi-tenant, missed-call→SMS→book, same-day invoice + Xero, digest, basic dashboard.
- **Phase C:** billing + self-serve onboarding + lead revival + pricing guard.
- **Phase D:** voice answering, full dashboard parity with the demo, MYOB, analytics.

The static demo (this site) is the sales asset that funds Phase A. Sell first, build second.